FLAIR Task Schematic

ICU Daily LTACH Discharge at 07:00 Local (Task T2)

FLAIR Benchmark

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Abstract

We specify the FLAIR daily ICU LTACH-discharge task: at every local 07:00 rounds anchor during the first ICU stay, predict whether the hospitalization will end with transfer to a Long-Term Acute Care Hospital (CLIF `discharge_category` = `Long Term Care Hospital (LTACH)`). The cohort is identical to Task T1 (all adult ICU hospitalizations, no `discharge_category` filter); only the label predicate differs. T1 and T2 are **not** a partition: a stay discharged Home is label-0 in both; a stay discharged Expired is label-1 in T1 and label-0 in T2. The shared pipeline is specified once in `icu_daily_mortality.tex` (Task T1); this schematic specifies only the T2 label rule and its design rationale.

1 Relationship to Task T1

The build pipeline of Task T2 is byte-identical to Task T1's: same cohort filters (adult, valid admission/discharge, ≥ 1 ICU ADT record, first ICU stay long enough for at least one 07:00 grid point), same pred_start, pred_stop, and decision-point grid, same 24h lookback, same temporal split. **Neither task filters on `discharge_category` at the cohort stage.** The only difference is the final label rule. The full operational definition, cohort details, and step-by-step build logic are in `icu_daily_mortality.tex` (Task T1); the flowchart of Figure 1 is shared.

2 Label rule (Task T2 only)

Let $\texttt{disp}_i = \texttt{lower}(\texttt{strip}(\texttt{discharge_category}_i))$. For d in the daily 07:00 decision grid D_i:

$$y_i(d) \;=\; \mathbb{1}[\,\texttt{disp}_i \text{ matches regex } \texttt{ltach|long.?term}\,].$$

- The label is constant across all rows of H_i (the eventual disposition is a single hospitalization-level fact broadcast to every grid point).
- A null $\texttt{disp}_i$ (i.e., `discharge_category` null in the source) or any non-matching value (`Expired`, `Hospice`, `Home`, `Skilled Nursing Facility (SNF)`, `Acute Inpatient Rehab Facility`, `Against Medical Advice (AMA)`, `Other`, `Still Admitted`, ...) yields $y_i(d) = 0$.

- Because Tasks T1 and T2 share the same unconditional cohort and have non-overlapping label predicates, $\texttt{label_mortality}_i + \texttt{label_ltach}_i \leq 1$ but **not** necessarily $= 1$: a Home discharge is 0 in both, an Expired discharge is 1 in T1 / 0 in T2, and an LTACH discharge is 0 in T1 / 1 in T2.

3 CLIF tables and categories used

The CLIF surface for Task T2 is identical to Task T1: the same three tables (`hospitalization`, `patient`, `adt`) and the same cohort filters. The ADT filter restricts to `location_category` = `"icu"` (case-insensitive); `patient` is join-only; `discharge_category` is read from `hospitalization` for the *label* only — no cohort filter is applied on it. See the full table and per-table column inventory in the sibling document `icu_daily_mortality.tex` (§CLIF tables and categories used). Tasks T1 and T2 differ only in the final label predicate, not in which CLIF data they touch.

4 Algorithmic flow (shared with Task T1)

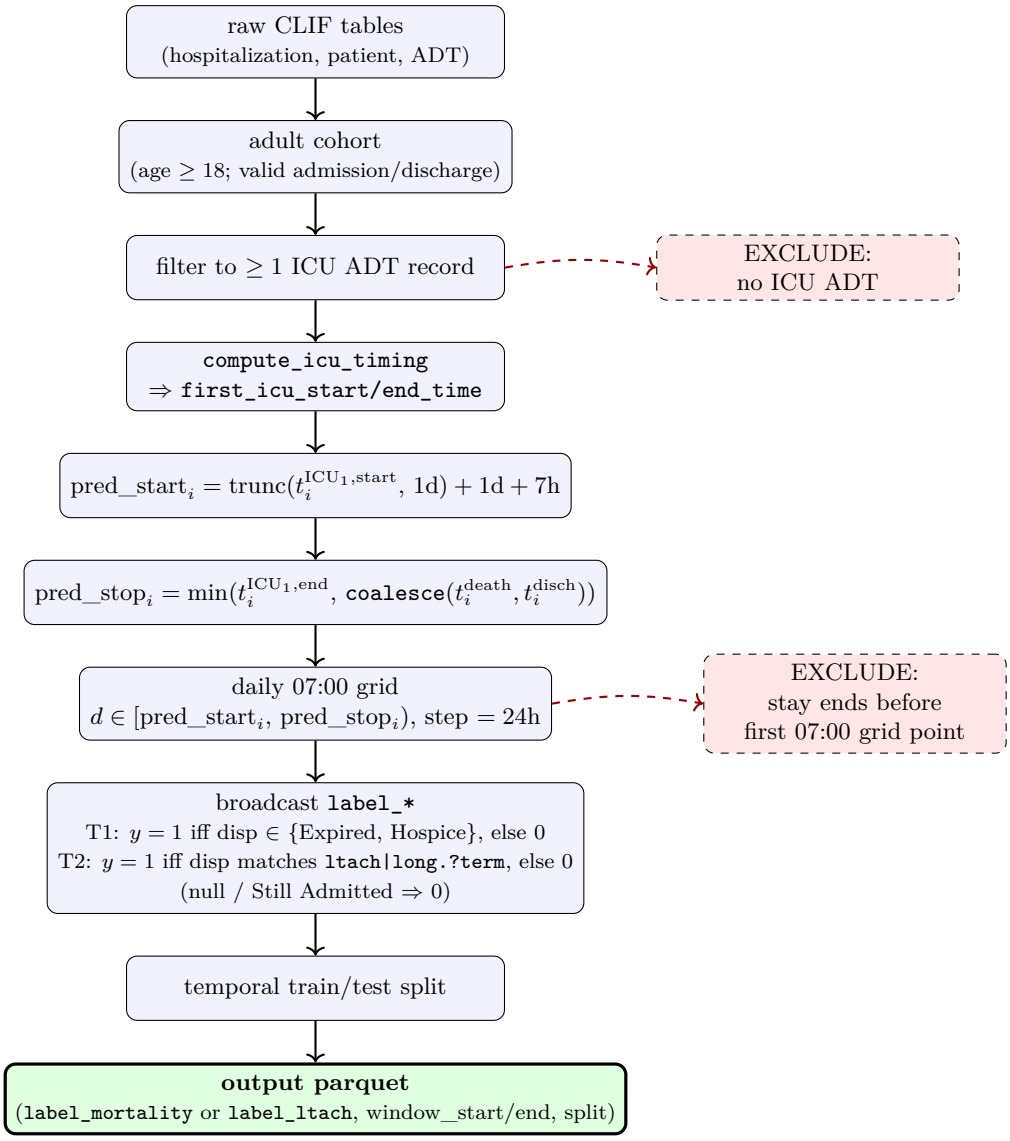


Figure 1: Build pipeline shared by Tasks T1 and T2. The label box at the bottom is parameterized: T2 emits $y = 1$ iff $\texttt{disp}_\text{i}$ matches `ltach|long.?term`. Everything else — cohort filters, ICU timing, pred_start / pred_stop, the daily 07:00 grid, the 24h lookback, the temporal split, the parquet schema — is identical to Task T1.

5 Design choices and rationale

Why a separate schematic for a different label on the same cohort. A reader scanning the supplement should find one document per FLAIR task, with its own META, label rule, and prevalence numbers. The build pipeline being shared is a convenience of implementation, not a property of the task definition: a future site whose `discharge_category` taxonomy splits LTACH from other long-term destinations more finely could change Task T2's regex without affecting

Task T1.

Prevalence and class imbalance. LTACH transfer is a rare disposition in the absolute sense: across all adult ICU stays it sits in the low single digits per-row positive rate. Class imbalance is therefore the dominant modelling challenge, and AUPRC is the primary headline metric. AUROC is reported but is known to be optimistic on heavily imbalanced binary tasks.

Why LTACH is the operationally interesting label. LTACH transfer is a planned, resource-intensive disposition decision that triggers care-coordination work several days before discharge (Kahn et al., 2010). A morning prediction of LTACH-bound trajectory can drive social-work referrals, family conversations, and family-meeting timing.

Why an unconditional cohort. Earlier drafts conditioned the cohort on the union of LTACH and in-hospital mortality, manufacturing a balanced binary against T1. That design has been retired (see Task T1 schematic, §Design choices). The unconditional cohort makes the per-row positive rate interpretable as the prevalence of LTACH-bound trajectories among all ICU survivors-and-non-survivors, and matches the standard ICU-outcome-prediction setup.

6 Out of scope

- All items listed in `icu_daily_mortality.tex` (Task T1) §*Out of scope* carry over.
- Finer-grained LTACH-vs-skilled-long-term-care distinctions within the LTACH-positive class.

References

Jeremy M. Kahn, Nicole M. Benson, Dina Appleby, Shannon S. Carson, and Theodore J. Iwashyna. Long-term acute care hospital utilization after critical illness. *JAMA*, 303(22):2253–2259, 2010. doi: 10.1001/jama.2010.761.